

(TR-104) PYTHON WITH AI/ML

Training Day 93 Report

Date: 20 May 2026

Topic Covered: Face Recognition, Voice Recognition, Streamlit Optimization & Project Documentation

The ninety-third day of training focused on analyzing face recognition and voice recognition workflows, studying AI model integration techniques, exploring Streamlit optimization and deployment concepts, and preparing project documentation for academic submission.

- Refactored the face recognition module using professional and meaningful naming conventions to improve code readability and maintainability.
- Studied dlib-based face detection and facial embedding generation workflows.
- Analyzed the generation and usage of 128-dimensional facial embeddings for student recognition.
- Reviewed SVM-based classification techniques used for attendance prediction and student identification.
- Studied Euclidean distance-based similarity matching and threshold-based face recognition mechanisms.
- Verified attendance marking workflows and unknown face detection handling.
- Explored voice recognition systems using Resemblyzer for speaker embedding generation.
- Analyzed speaker identification workflows based on cosine similarity matching techniques.
- Reviewed bulk audio processing and multi-speaker recognition methodologies.
- Refactored voice recognition modules to improve modularity, organization, and code clarity.
- Studied audio segmentation techniques and speaker verification threshold handling.
- Reviewed exception handling and fallback mechanisms for voice recognition and audio processing operations.

- Explored Streamlit caching concepts using resource caching techniques for optimized model loading.
- Analyzed AI model loading workflows and resource management strategies in Streamlit applications.
- Studied frontend-backend synchronization, session handling, and workflow execution during attendance processing.
- Explored deployment workflows for Streamlit-based AI applications and reviewed production-level project structures.
- Analyzed the integration of AI models, databases, and frontend components within complete application workflows.
- Prepared daily diary documentation and organized project implementation records for college submission.
- Reviewed project documentation related to NLP workflows, preprocessing techniques, model training concepts, and implementation details.
- Revised concepts related to face recognition, facial embeddings, speaker recognition, voice embeddings, SVM classification, Streamlit optimization, deployment workflows, and modular AI application development.

Overall, the training enhanced understanding of face recognition systems, voice recognition techniques, AI model integration, Streamlit optimization, deployment workflows, and technical documentation practices.